

6.5 Classwork: Quadratic Functions and Symmetry — Markscheme

1. The diagram shows the graph of a quadratic function with x -intercepts at $(1, 0)$ and $(5, 0)$.
 - (a) Find the equation of the axis of symmetry. [1]
 - (b) Write the equation of the parabola in factored form, $y = (x - p)(x - q)$. [1]
 - (c) Expand to show that the equation is $y = x^2 - 6x + 5$. [2]
 - (d) Find the coordinates of the vertex. [2]
 - (e) Write down the y -intercept. [1]

Solutions

(a) $x = \frac{1 + 5}{2} = 3$ (A1)

The axis of symmetry is $x = 3$.

(b) $y = (x - 1)(x - 5)$ (A1)

(c) $y = (x - 1)(x - 5)$
 $= x^2 - 5x - x + 5$ (M1)

$= x^2 - 6x + 5$ (AG)

(d) $y = (3)^2 - 6(3) + 5 = 9 - 18 + 5 = -4$ (M1)

Vertex: $(3, -4)$ (A1)

(e) y -intercept: $x = 0$, $y = 0 - 0 + 5 = 5$, so $(0, 5)$ (A1)

2. The diagram shows the graph of $g(x) = -(x - 2)^2 + 9$.

(a) Write down the coordinates of the vertex. [1]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that $g(x) = -x^2 + 4x + 5$. [2]

(d) Solve $g(x) = 0$ to find the x -intercepts. [3]

Solutions

(a) Vertex: $(2, 9)$ (A1)

(b) $x = 2$ (A1)

(c) $g(x) = -(x - 2)^2 + 9$
 $= -(x^2 - 4x + 4) + 9$ (M1)

$= -x^2 + 4x - 4 + 9$
 $= -x^2 + 4x + 5$ (AG)

(d) $-x^2 + 4x + 5 = 0$
 $x^2 - 4x - 5 = 0$ (multiply both sides by -1) (M1)

$(x - 5)(x + 1) = 0$ (M1)

$x = 5$ or $x = -1$ (A1)

The x -intercepts are $(-1, 0)$ and $(5, 0)$.

3. The diagram shows the graph of $h(x) = x^2 - 2x - 3$.
- (a) Write $h(x)$ in factored form, $(x - p)(x - q)$. [2]
 - (b) Write down the x -intercepts of the parabola. [1]
 - (c) Find the equation of the axis of symmetry. [1]
 - (d) Find the coordinates of the vertex. [2]
 - (e) Write $h(x)$ in vertex form, $(x - h)^2 + k$. [1]

Solutions

- (a) Find two numbers that multiply to -3 and add to -2 : (M1)
 $h(x) = (x + 1)(x - 3)$ (A1)
- (b) x -intercepts: $(-1, 0)$ and $(3, 0)$ (A1)
- (c) $x = \frac{-1 + 3}{2} = 1$ (A1)
- (d) $h(1) = (1)^2 - 2(1) - 3 = 1 - 2 - 3 = -4$ (M1)
Vertex: $(1, -4)$ (A1)
- (e) $(x - 1)^2 - 4$ (A1)

4. Consider the quadratic function $p(x) = -x(x - 4)$ with domain $-1 \leq x \leq 5$.
- (a) Write down the x -intercepts. [1]
 - (b) Find the equation of the axis of symmetry. [1]
 - (c) Find the coordinates of the vertex. [2]
 - (d) Expand to write $p(x)$ in the form $ax^2 + bx + c$. [1]
 - (e) On the grid below, sketch the graph of $p(x)$. Label the vertex and x -intercepts. [2]
 - (f) Write down the range of the function over the given domain. [1]

Solutions

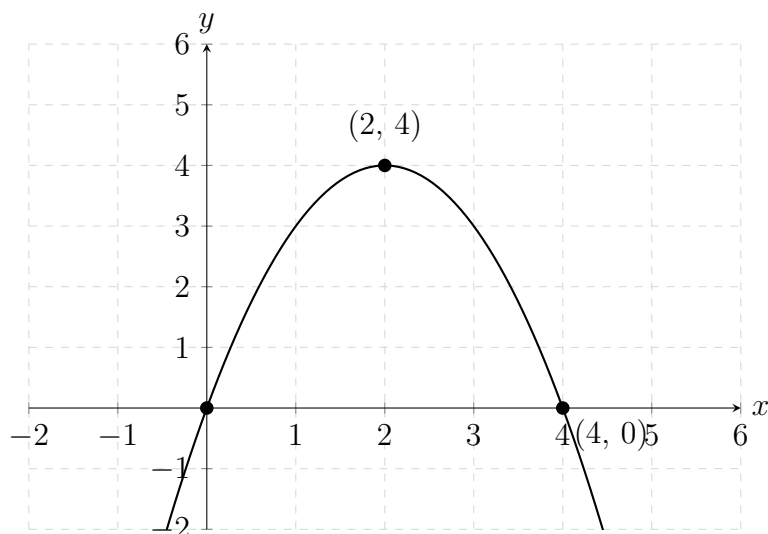
(a) x -intercepts: $(0, 0)$ and $(4, 0)$ (A1)

(b) $x = \frac{0 + 4}{2} = 2$ (A1)

(c) $p(2) = -2(2 - 4) = -2(-2) = 4$. Vertex: $(2, 4)$ (M1)(A1)

(d) $p(x) = -x^2 + 4x$ (A1)

(e) Graph: downward-opening parabola through $(0, 0)$ and $(4, 0)$ with vertex at $(2, 4)$.
(A1) shape, (A1) key points



(f) Over the domain $-1 \leq x \leq 5$, the vertex gives the maximum $p(2) = 4$, and the endpoints give the minimum: $p(-1) = -(-1)(-5) = -5$ and $p(5) = -(5)(1) = -5$.
(A1)

Range: $-5 \leq p(x) \leq 4$.